POLIMI GRADUATE SCHOOL OF MANAGEMENT

Master in Business Analytics and Data Science · Final Project Work

Keeping LLMs Current

From NLP to Agentic Search on Serie A

Alexandre Benjamin Cambier Tutor: Prof. Andrea Mor Academic Year 2025–2026

OPENING

A Question That Looks Easy

"Who coaches the team directly above Cremonese in the Serie A table right now?"

A **Multi-Hop** Question: Two Facts in a Chain, and **the Club in the Middle Is Never Named**.

OPENING

Same Question, Different Answer

"Who coaches the team directly above Cremonese in the Serie A table right now?"

SEP 1

1 September 2025

After Matchday 2

POS	CLUB	PTS
1	Juventus	6
2	Napoli	6
3	Cremonese	6
4	Roma	6

1 One Place Up

Napoli and Juventus Are Level on Points, Goal Difference and Goals: If Juventus Is Above, the Answer Is Igor Tudor.

2 Its Coach

COACH OF NAPOLI
Antonio Conte

≠

MAY 25

25 May 2026

The Season Is Over

POS	CLUB	PTS
16	Genoa	41
17	Lecce	38
18	Cremonese Relegated	34
19	Verona	21

1 One Place Up

2 Its Coach

COACH OF LECCE
Eusebio Di Francesco

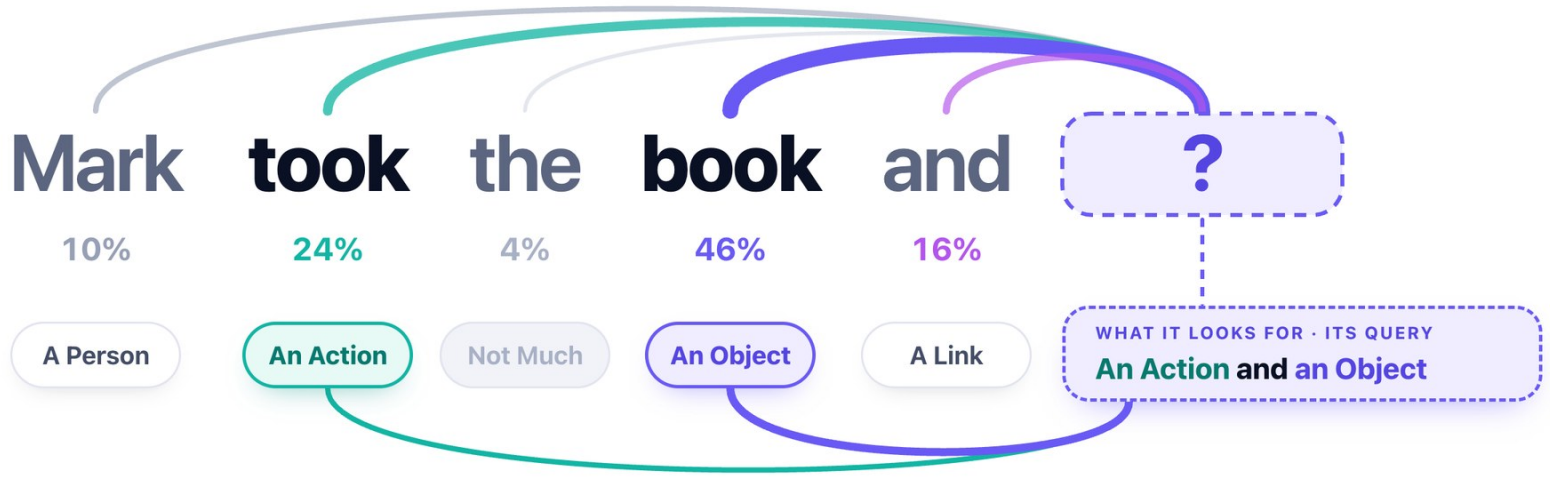
Who Should Decide How to Search for Fresh Facts? The Engineer or The Model

THEORY

How a Language Model Writes: Attention

1 ATTENTION To Guess the Blank, It Weighs Each Word so Far

They Add Up to **100%**



Best Match, Most Weight 46% to “book”, 24% to “took” • Keys and Queries Are Learned in Training

THEORY

How a Language Model Writes: Attention

1 **ATTENTION** To Guess the Blank, It Weighs Each Word so Far

They Add Up to **100%**

2 IT BLENDS THE WORDS BY THOSE WEIGHTS

Mostly **"book"**, Then **"took"**: Someone Took a Book.

3 IT SCORES EVERY WORD IT KNOWS AGAINST THE SUMMARY

4 It Writes the Likeliest, **"read"**, Adds It to the Text, and Does It All Again for the Next Word

THEORY

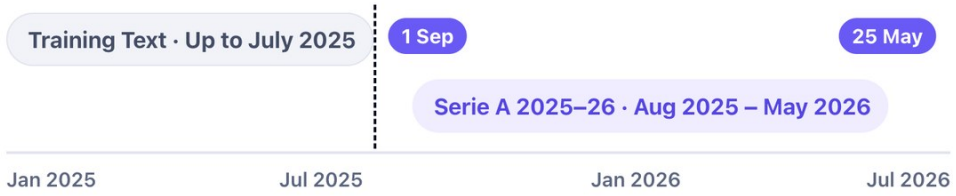
How a Language Model Writes, and Its Two Limits

1 ATTENTION To Guess the Blank, It Weighs Each Word so Far

They Add Up to **100%**

What It Learned Is **Frozen in Its Weights**

LIMIT 1 · KNOWLEDGE CUTOFF



239 of 240

Answers Were "unknown" From Claude Haiku 4.5 Alone; the One Answer It Gave Was ✕ **Wrong**

It Picks a **Likely** Word, Not a **Checked** One

LIMIT 2 · HALLUCINATION

The Most Likely Word Is Not Always True

6

Court Rulings **Invented by ChatGPT**, Cited by Lawyers in **Mata v. Avianca** (2023)

THEORY

The Standard Fix: Retrieval-Augmented Generation

1 ? **Question**

→

2 **Search**
A Library of Documents
UPDATED ANYTIME

→

3 **Best Passages**
Top 5

→

4 **The LLM Writes**
From the Passages

→

5 **Answer**

TWO MEMORIES · LEWIS ET AL., 2020

Update the Library, Not the Model

Parametric
The Model's Weights, Learned in Training.
FIXED

Non-Parametric
A Library of Documents,
Searched at Question Time.
UPDATED ANYTIME

FRESHPROMPT · VU ET AL., 2024

+32.6 / +49.0

Points for GPT-4 on **FreshQA**, a Benchmark of Changing Facts, Under Relaxed / Strict Grading, Just by Adding Fresh Search Results.

THEORY

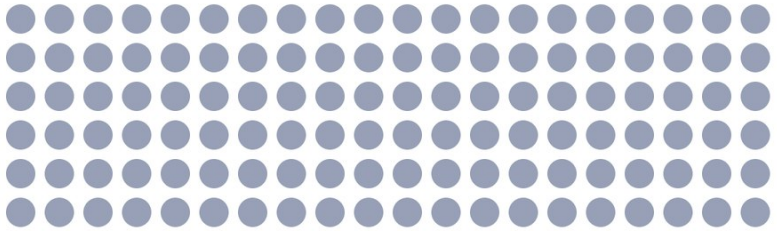
The Test: 120 Questions, Asked on Two Dates

A TEMPLATE: ONE SENTENCE, ONE BLANK · "COACH OF THE CLUB ABOVE", MULTI-HOP

"Who coaches the team directly above Hellas Verona in the Serie A table right now?"

17 Templates
Each Written 4–5 Ways

Each Blank Filled
4–9 Times
With a Club or a Position



120 Questions
One for Every Way of Filling a Blank

EACH QUESTION, ASKED TWICE · THE RIGHT ANSWER READ FROM THAT DATE'S TABLE

SEP 1 — **1 September 2025** — After Matchday 2 → COACH OF LECCE **Eusebio Di Francesco**

MAY 25 — **25 May 2026** — The Season Is Over → COACH OF CREMONESE **Marco Giampaolo**

THEORY

The Test: 120 Questions, Asked on Two Dates

17 TEMPLATES → 120 QUESTIONS · EACH ASKED ON 1 SEP 2025 AND ON 25 MAY 2026

"Who coaches the team directly above Hellas Verona in the Serie A table right now?"

STATIC · 40 QUESTIONS

About Last Season

ITS 5 TEMPLATES

Final Position · Final Place · Final Points · Final Goals · Final-Day Coach

TEMPLATE "FINAL POSITION"

"Which team finished 18th in Serie A in 2024-25?"

Both Dates **Empoli**

DYNAMIC · 40 QUESTIONS

About Now

ITS 5 TEMPLATES

Coach Now · Team at a Position · Position Now · Points Now · Goals Now

TEMPLATE "COACH NOW"

"Who is the head coach of Fiorentina right now?"

1 Sep **Stefano Pioli**

25 May **Paolo Vanoli**

MULTI-HOP · 40 QUESTIONS

Two Facts in a Chain

ITS 7 TEMPLATES

Coach of the Club Above · Club Above · Coach at a Position · Gap to the Leader · Gap Between Two Clubs · Last Season's Club, Now · Last Season's Final-Day Coach

TEMPLATE "COACH OF THE CLUB ABOVE"

"Who coaches the team directly above Sassuolo in the Serie A table right now?"

1 Sep **Marco Baroni**

25 May **Kosta Runjaić**

THEORY

The Library: 22 Pages per Date, Cut Into Passages

22 WIKIPEDIA PAGES

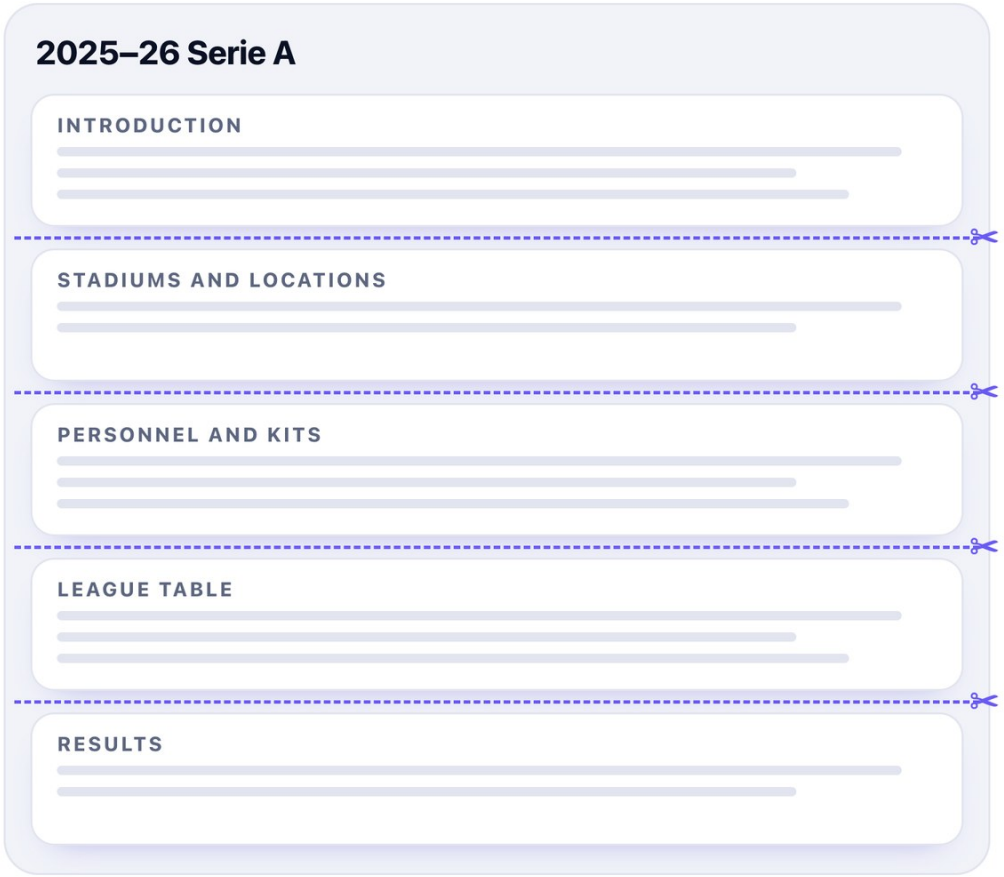
Each Saved on 1 Sep 2025 25 May 2026

THE 20 CLUBS

2025–26 Serie A
This Season

Last Season (2024–25): a Trap

ONE PAGE, CUT AT ITS SECTIONS



THEORY

The Library: 22 Pages per Date, Cut Into Passages

22 WIKIPEDIA PAGES

Each Saved on 1 Sep 2025 25 May 2026

THE 20 CLUBS

2025–26 Serie A
This Season

Last Season (2024–25): a Trap

PASSAGES · ABOUT 140 WORDS, EACH WITH A LABEL

2025–26 SERIE A · 2025-26 · INTRODUCTION

`The 2025–26 Serie A … was the 124th season of top-tier Italian football …`

2025–26 SERIE A · 2025-26 · STADIUMS AND LOCATIONS

`Team: Lecce · Location: Lecce · Stadium: Stadio Via del Mare–Ettore Giardiniero …`

2025–26 SERIE A · 2025-26 · PERSONNEL AND KITS

`Team: Lecce · Chairman: Saverio Sticchi Damiani · Manager: Eusebio Di Francesco …`

2025–26 SERIE A · 2025-26 · LEAGUE TABLE · D1 · Shortened

`Pos: 17 · Team: Lecce · Pts: 38 · Pos: 18 · Team: Cremonese (R) · Pts: 34 …`

2025–26 SERIE A · 2025-26 · RESULTS

`Home \ Away: Lecce · ATA: 0–3 · BOL: 2–2 · CAG: 1–2 · COM: 0–3 · CRE: 2–1 …`

946 Passages on 1 Sep 2025 **993** on 25 May 2026

5 A Search Returns the **5 Best Passages**

An Agent Can Also Read a **Whole Page**

THEORY

Search by Words: BM25

THE BM25 SCORE OF A PASSAGE D FOR A QUESTION Q · THESIS EQ. 2.2

$$\text{score}(q, d) = \sum_t \underbrace{\text{IDF}(t)}_{1} \times \underbrace{\frac{\text{tf}(t, d) \times (k_1 + 1)}{\text{tf}(t, d) + k_1 \times \underbrace{(1 - b + b \times |d| / \text{avgdl})}_{3}}}_{2}$$

1 How Rare the Word Is · 2 How Often It Appears, With a Ceiling · 3 d's Length Against the Average

d: A Passage, D1–D5 · Σ: Over the Question's Words t · tf: Times t Appears in d · |d|: Its Length in Words · avgdl: The Average Length · $k_1 = 1.5 \cdot b = 0.75$

1 IDF(t)

Rare Words Weigh More

	D1	D2	D3	D4	D5	WEIGHT
table	■	■				0.88
team	■	■			■	0.54
Cremonese	■			■	■	0.54
Serie	■	■			■	0.54
coaches, above						No Match

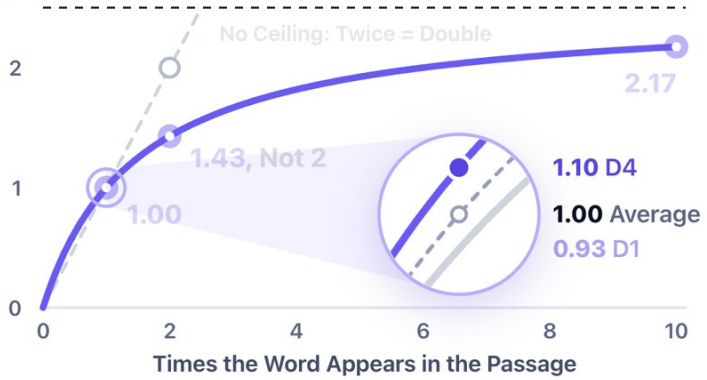
table: In 2 of 5 Passages → Rarer → **0.88**
team: In 3 of 5 → **0.54**

GIVES **One Weight per Question Word**

2 $\text{tf} \times (k_1 + 1) \div (\text{tf} + k_1 \times \ldots)$

Repeats Help Less and Less

2 What the Word Counts — Ceiling 2.5



2 For One "Cremonese" in D4, 13 Words

$$1 \times 2.5 \div (1 + 1.5 \times 0.85) = 1.10$$

3 $(1 - b + b \times |d| / \text{avgdl})$

Long Passages Score a Bit Lower

D4 Cremonese's Page — 13 words

"Cremonese", Once

3 $1 - 0.75 + 0.75 \times 13 / 16.2 = 0.85$

GIVES 3 Sits Inside 2: Longer Passage, Smaller 2

THEORY

Search by Words: One Passage, Worked Out

"Who coaches the team directly above Cremonese in the Serie A table right now?"

Demo · 5 Shortened Passages · 25 May

D1 **Today's Table** 2025–26 Serie A (2025-26) · League table

Pos: 17 · Team: Lecce · Pts: 38

Pos: 18 · Team: Cremonese (R) · Pts: 34

19 Words, a Bit Longer Than the Average, 16.2

Part ③ for D1 = 1.13

Question Word in D1	① IDF: How Rare		② The Fraction, With ③ Inside		Its Share
table	0.875 In 2 of 5 Passages	×	0.928 1 × 2.5 ÷ (1 + 1.5 × 1.13)	=	0.81
team ×2	0.539 In 3 of 5	×	1.353 2 × 2.5 ÷ (2 + 1.5 × 1.13)	=	0.73
Cremonese	0.539 In 3 of 5	×	0.928 1 × 2.5 ÷ (1 + 1.5 × 1.13)	=	0.50
Serie	0.539 In 3 of 5	×	0.928 1 × 2.5 ÷ (1 + 1.5 × 1.13)	=	0.50

Σ · D1's Score

0.81 0.73 0.50 0.50

2.54

THEORY

Search by Words: All Five Passages, Ranked

“Who coaches the team directly above Cremonese in the Serie A table right now?”

Real Run Every One of ~1,000 Passages Is Scored; **the Best 5** Go On

Demo · 25 May 5 Shortened Passages; **the Best 2** Go On

TOP 2

#1

D1 **Today’s Table** 2025–26 Serie A (2025-26) · League table

Pos: 17 · Team: Lecce · Pts: 38

Pos: 18 · Team: Cremonese (R) · Pts: 34

table 0.81 | team ×2 0.73 | Cremonese 0.50 | Serie 0.50

2.54

#2

D2 **Last Season’s Table** 2024–25 Serie A (2024-25) · League table

Pos: 17 · Team: Lecce · Pts: 34

Pos: 18 · Team: Empoli (R) · Pts: 31

table 0.81 | team ×2 0.73 | Serie 0.50

No “Cremonese”, Yet Second: **Last Season’s Table**

2.04

#3

D5 **Today’s Stadiums** 2025–26 Serie A (2025-26) · Stadiums and locations

Team: Cremonese · Location: Cremona

Stadium: Stadio Giovanni Zini

team 0.56 | Cremonese 0.56 | Serie 0.56

1.67

#4

D4 **Cremonese’s Page** US Cremonese (2025-26) · Introduction

Ground: Stadio Giovanni Zini

Head coach: Marco Giampaolo

Cremonese 0.59

0.59

#5

D3 **Lecce’s Page** US Lecce (2025-26) · Introduction

Ground: Stadio Via del mare

Head coach: Eusebio Di Francesco

No Question Word

The Answer, at 0: “coaches” ≠ “coach”, and Lecce Is Not in the Question

0.00

Why Several, Not One A Chained Answer Needs Facts From Different Passages: Here, **Today’s Table** and **a Coach’s Page**

THEORY

Classical NLP: Search by Words, Then a Reader Copies a Span

S1 · Classical NLP — ? Question — Aa Search by Words Best 5 of ~1,000 — A Reader Copies a Span — ✓ Answer

"Who is the head coach of Fiorentina right now?"

Answer: **Paolo Vanoli** ✓ Right

ITS BEST 5 BY WORDS	ITS BEST SPAN IN EACH	SPAN SCORE VS "NOT HERE" SCORE	↻ RE-RUN OFFLINE
1 Atalanta BC — Managers	START Raffaele Palladino END	3.2 — Not Here 9.3	Not Here Wins
2 2024–25 Serie A — League Table	START Monza END	2.3 — Not Here 8.0	Not Here Wins
3 **ACF Fiorentina** — Introduction	START **Paolo Vanoli** END	**16.5** — Not Here 6.0	**Surest · Copied**
4 **ACF Fiorentina** — Della Valle Era	START **Cesare Prandelli** END	**14.9** — Not Here 6.3	Less Sure
5 **Bologna FC 1909** — Climbing Down and Back Up	START **Ulivieri** END	**11.9** — Not Here 6.1	Less Sure

Copies, Never Writes
Only Words Already in a Passage

One Piece of One Passage
It Cannot Join Two

None Beats "Not Here"?
It Says "Unknown", as S1 Will on Our Question

THEORY

Search by Meaning: A Text Becomes 384 Numbers

“Who coaches the team directly above Cremonese in the Serie A table right now?” → **Encoder** bge-small →

[`−0.003` `−0.031` `−0.010` `−0.073` `0.039` `0.001` `0.007` `0.026` … 376 more]

Its **Embedding**: No Single Number Means Anything; Together They Place the Text **Where Similar Meanings Sit Close**.

Sparse · Counting Words

One Slot for **Every Word** of the Vocabulary (a Few Shown): Almost All 0

	club	**coach**	league	**manager**	match	season	table	team	
“coach”	0	**1**	0	0	0	0	0	0	…
“manager”	0	0	0	**1**	0	0	0	0	…

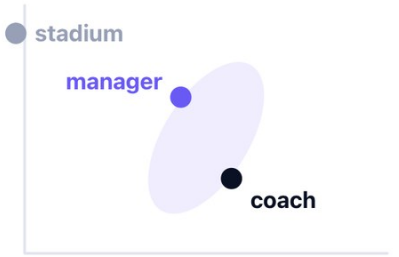
✕ No Slot in Common: They Look **Unrelated**

Dense · An Embedding

Two Numbers per Text (Real: 384)

A Short List of Numbers, **Every One Used**

“coach”	0.94	0.34
“manager”	0.71	0.71



✓ Close Numbers: **Close Meaning**, With No Word in Common

THEORY

Search by Meaning: A Text Becomes 384 Numbers

“Who coaches the team directly above Cremonese in the Serie A table right now?” → **Encoder** bge-small →

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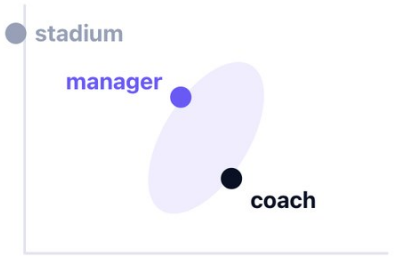
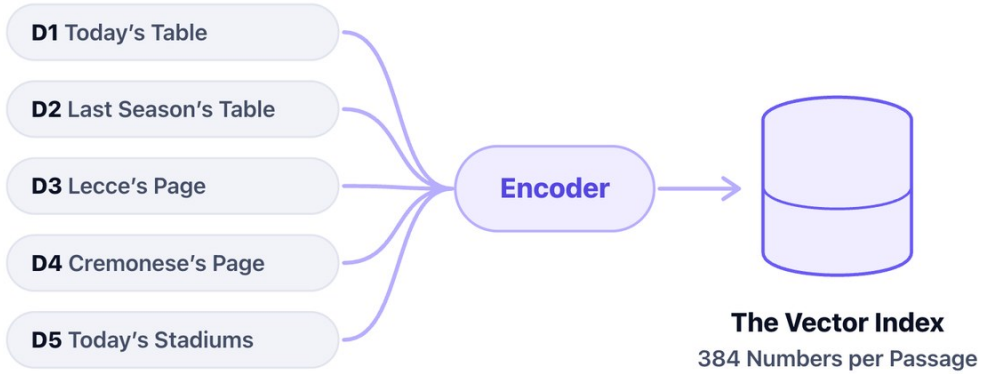


Close Numbers: **Close Meaning**, With No Word in Common

Stored in Advance: The Vector Index

Every Passage Goes Through the **Same Encoder**, Once, Before Any Question



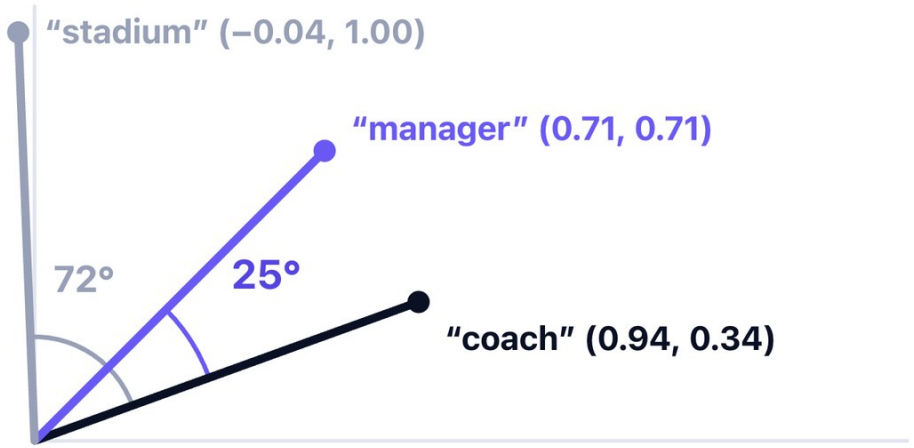
The Vector Index
384 Numbers per Passage

THEORY

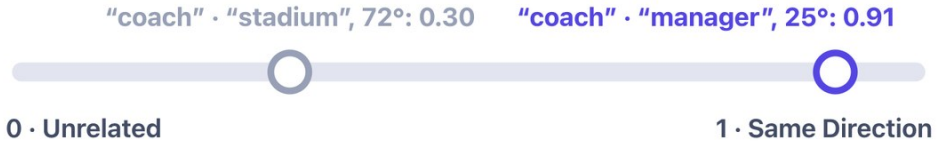
Search by Meaning: Compare by the Angle

EACH TEXT IS AN ARROW

Two Numbers per Text (Real: 384)



Small Angle: Similar Meaning · **Wide Angle**: Different Meaning



THE COSINE: THE ANGLE AS ONE NUMBER

$$\cos(a, b) = \frac{a \cdot b}{\|a\| \times \|b\|}$$

(1) (2)

1 Top: Multiply Pair by Pair, Then Add Up

Here, Two Numbers `0.94 × 0.71 + 0.34 × 0.71 = 0.91`

2 Bottom: The Two Lengths, So Only Direction Counts

Here `‖coach‖ = 1.00 · ‖manager‖ = 1.00`

cos = 0.91 ÷ (1.00 × 1.00) = **0.91**

THEORY

Search by Meaning: The Same Five Passages

"Who coaches the team directly above Cremonese in the Serie A table right now?"

Demo · The Same Five as BM25 No Pre-Selection: Each One Gets a Score, **Its Cosine With the Question**; Then They Are Sorted

#1
D4 **Cremonese's Page** US Cremonese (2025-26) · Introduction
Ground: Stadio Giovanni Zini
Head coach: Marco Giampaolo
1 **Wrong Club** **0.800**

#2
D1 **Today's Table** 2025–26 Serie A (2025-26) · League table
Pos: 17 · Team: Lecce · Pts: 38
Pos: 18 · Team: Cremonese (R) · Pts: 34
1 **0.781**

TOP 2

#3
D5 **Today's Stadiums** 2025–26 Serie A (2025-26) · Stadiums and locations
Team: Cremonese · Location: Cremona
Stadium: Stadio Giovanni Zini
1 **0.779**

#4
D3 **Lecce's Page** US Lecce (2025-26) · Introduction
Ground: Stadio Via del mare
Head coach: Eusebio Di Francesco
1 **The Answer · Only Fourth** **0.736**

#5
D2 **Last Season's Table** 2024–25 Serie A (2024-25) · League table
Pos: 17 · Team: Lecce · Pts: 34
Pos: 18 · Team: Empoli (R) · Pts: 31
1 **Last Season** **0.724**

All Five: 0.72 – 0.80 By Meaning, Every Serie A Passage **Looks Alike**; and the Question **Never Names Lecce**

THEORY

Advanced RAG: Better Parts, Still One Search

"Who coaches the team directly above Cremonese in the Serie A table right now?"

Demo · 5 Passages · Best 2 Kept

1. **Two Searches** — Words + Meaning
2. **Merge Rankings** — Rank Fusion
3. **Season Filter** — Drops 2024–25
4. **Re-ranker** — A Cross-Encoder
5. **The LLM Writes** — From the Top 2

BY WORDS · BM25

#1	D1	Today's Table	2.54
#2	D2	Last Season's Table	2.04
#3	D5	Today's Stadiums	1.67
#4	D4	Cremonese's Page	0.59
–	D3	No Shared Word	

BY MEANING · BI-ENCODER

#1	D4	Cremonese's Page	0.800
#2	D1	Today's Table	0.781
#3	D5	Today's Stadiums	0.779
#4	D3	Lecce's Page	0.736
#5	D2	Last Season's Table	0.724

THE REAL RUN, AND THIS DEMO

	Passages		Kept by Words + Meaning		To the LLM
Real Run	~1,000	→	20 + 20	→	5
Demo	5	→	4 + 5	→	2

WHY TWO SEARCHES

- Aa **Words** Catch Exact Names and Numbers
- ≈ **Meaning** Catches Synonyms

THEORY

Advanced RAG: Better Parts, Still One Search

"Who coaches the team directly above Cremonese in the Serie A table right now?"

Demo · 5 Passages · Best 2 Kept

1 **Two Searches** — Words + Meaning
→ 2 **Merge Rankings** — Rank Fusion
→ 3 **Season Filter** — Drops 2024–25
→ 4 **Re-ranker** — A Cross-Encoder
→ 5 **The LLM Writes** — From the Top 2

BY WORDS · BM25				BY MEANING · BI-ENCODER				MERGED · RANK FUSION		
#1	D1	Today's Table	2.54	#1	D4	Cremonese's Page	0.800	#1	D1	1/61 + 1/62 = 0.0325
#2	D2	Last Season's Table	2.04	#2	D1	Today's Table	0.781	#2	D4	1/64 + 1/61 = 0.0320
#3	D5	Today's Stadiums	1.67	#3	D5	Today's Stadiums	0.779	#3	D5	1/63 + 1/63 = 0.0317
#4	D4	Cremonese's Page	0.59	#4	D3	Lecce's Page	0.736	#4	D2	1/62 + 1/65 = 0.0315
–	D3	No Shared Word		#5	D2	Last Season's Table	0.724	#5	D3	– + 1/64 = 0.0156

RECIPROCAL RANK FUSION

$$RRF(d) = \frac{1}{60 + rank_1} + \frac{1}{60 + rank_2}$$

$rank_1$: d's Place by Words · $rank_2$: by Meaning

D1: First by Words, Second by Meaning
`1/61 + 1/62 = 0.0325`

Ranks, Not Scores: the Two Scales Differ.

High in Both Lists Wins. D3, in One List Only, Gets Half.

THEORY

Advanced RAG: Better Parts, Still One Search

"Who coaches the team directly above Cremonese in the Serie A table right now?"

Demo · 5 Passages · Best 2 Kept

1. **Two Searches** — Words + Meaning
2. **Merge Rankings** — Rank Fusion
3. **Season Filter** — Drops 2024–25
4. **Re-ranker** — A Cross-Encoder
5. **The LLM Writes** — From the Top 2

AFTER THE SEASON FILTER

#1	D1	1/61 + 1/62 = 0.0325
#2	D4	1/64 + 1/61 = 0.0320
#3	D5	1/63 + 1/63 = 0.0317
#4	D3	— + 1/64 = 0.0156

DROPPED: LAST SEASON

× ~~D2~~ ~~Last Season's Table~~

HOW: EVERY PASSAGE CARRIES A LABEL

The Label Names the Passage's Page, **Its Season** and Its Section; the Search Reads It With the Text.

D1	2025–26 Serie A (2025–26) · League table	✓ Kept
D2	~~2024–25 Serie A (2024–25) · League table~~	× Dropped

The Rule A Question About **Now** → Keep Only Labels With (2025–26)

Only S3	The Engineer Wrote This Rule
S2	No Filter: One Plain Search
S4–S5+	The Agents Choose the Season

THEORY

Advanced RAG: Better Parts, Still One Search

"Who coaches the team directly above Cremonese in the Serie A table right now?"

Demo · 5 Passages · Best 2 Kept

1. **Two Searches** — Words + Meaning
2. **Merge Rankings** — Rank Fusion
3. **Season Filter** — Drops 2024–25
4. **Re-ranker** — A Cross-Encoder
5. **The LLM Writes** — From the Top 2

THE SHORTLIST, TO RE-RANK

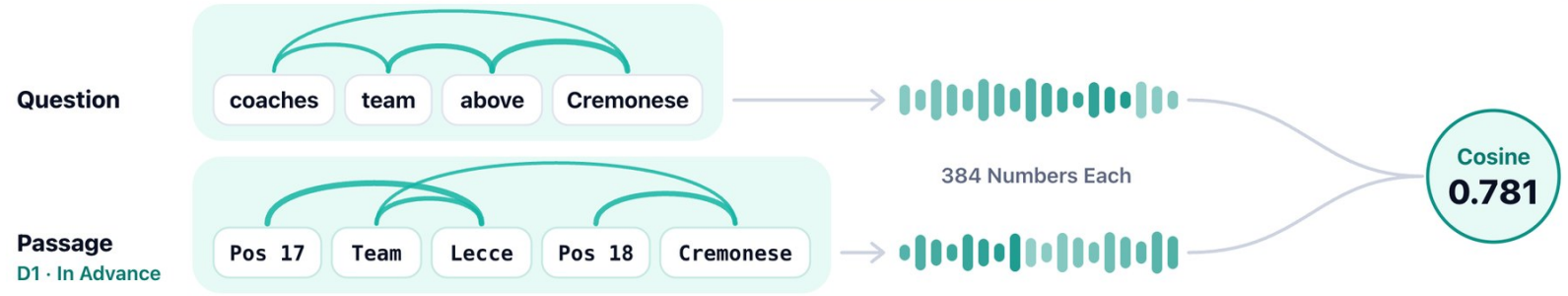
#1	D1	1/61 + 1/62 = 0.0325
#2	D4	1/64 + 1/61 = 0.0320
#3	D5	1/63 + 1/63 = 0.0317
#4	D3	– + 1/64 = 0.0156

Bi-Encoder

Our Search by Meaning · Every Passage

Attention Within Each Text

MODEL bge-small-en-v1.5

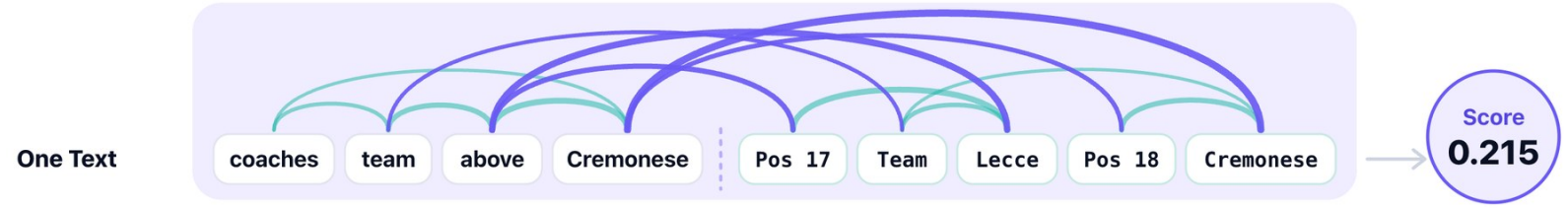


Cross-Encoder

The Re-ranker · Only the Top 20

Attention Across the Two Texts

MODEL bge-reranker-v2-m3



Question and Passage, Read Together

No Vectors, No Cosine

THEORY

Advanced RAG: Better Parts, Still One Search

"Who coaches the team directly above Cremonese in the Serie A table right now?"

Demo · 5 Passages · Best 2 Kept

1. **Two Searches** — Words + Meaning
2. **Merge Rankings** — Rank Fusion
3. **Season Filter** — Drops 2024–25
4. **Re-ranker** — A Cross-Encoder
5. **The LLM Writes** — From the Top 2

TOP 2 GO TO THE LLM

#1	D1	Today's Table	0.215
#2	D4	Cremonese's Page	0.061
#3	D3	Lecce's Page	0.036
#4	D5	Today's Stadiums	0.002

THE LLM WRITES FROM THE TOP 2

Example Reply

READS: D1 Today's Table, D4 Cremonese's Page

NOT READ: D3 Lecce's Page — Ranked #3

S3

The passages give the table and Cremonese's coach, but not Lecce's.

× unknown

Lecce Is Never Named in the Question, so **Its Page Never Reaches the Top**.

THEORY

Seven Systems, From the Engineer to the Model

The Real Systems · ~1,000 Passages per Date

NO ONE SEARCHES

S0 · Closed Book
No Search

Question (The Words as Asked) → Its Memory (Text Up to July 2025) → Answer

FIXED PIPELINES · THE ENGINEER FIXES THE STEPS

S1 · Classical NLP
Searches Once

Question → Search by Words (Best 5 of ~1,000) → A Reader (Copies a Span, No LLM) → Answer

S2 · Naive RAG
Searches Once

Question → Search by Meaning (Best 5 of ~1,000) → Haiku 4.5 Writes (From the Best 5) → Answer

S3 · Advanced RAG
Searches Once, Two Lists

Question → Words + Meaning (Best 20 + 20 of ~1,000) → Merge (One List) → Season Filter (Only 2025–26) → Re-ranker (Keeps the Best 5) → Haiku 4.5 Writes (From the Best 5) → Answer

AGENTS · THE MODEL DECIDES EACH NEXT STEP · UP TO 8 TOOL CALLS · EACH SEARCH RETURNS THE BEST 5

S4 · Agentic RAG

Question → Haiku 4.5 Decides ⇄ Search by Meaning, Read a Whole Page → Answer

S5 · Agentic Search

Question → Haiku 4.5 Decides ⇄ Search by Words, Read a Whole Page → Answer

S5+ · Agent + Tables

Question → Haiku 4.5 Decides ⇄ Search by Words, Read a Whole Page, Ask the Official Table → Answer

THEORY

Agents: The Model Takes Control

THE REACT LOOP · YAO ET AL., 2023

Think, **Act** With a Tool, **Observe** the Result

AT EVERY STEP, THE MODEL DECIDES

1. **Search, or Answer Now?**
2. **Which Words, With Which Tool?**
3. **Search Again, or Read a Whole Page?**

A Pipeline
The Same Fixed Steps
for Every Question

An Agent
Picks Each Next Step,
Up to **8 Tool Calls**

S4 **S4 · Agentic RAG**
- ≈ Search by Meaning
- ▤ Read a Whole Page

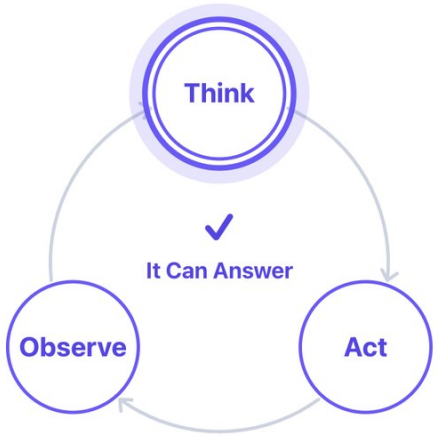
S5 **S5 · Agentic Search**
- Aa Search by Words
- ▤ Read a Whole Page

S5+ **S5+ · Agent + Tables**
- Aa Search by Words
- ▤ Read a Whole Page
- ▦ Ask the Official Table

THEORY

Agents: The Model Takes Control

THE REACT LOOP · YAO ET AL., 2023



Think, **Act** With a Tool, **Observe** the Result

S4

S4 · Agentic RAG

≈ Search by Meaning

▤ Read a Whole Page

S4 ON OUR QUESTION · Example Run · Five Demo Passages

CYCLE 1

THINK — I need the club just above Cremonese.

ACT — ≈ Search by Meaning — `Serie A standings 2025–26 Cremonese position`

OBSERVE — **D1** Today's Table

```
Pos: 17 · Team: Lecce · Pts: 38
Pos: 18 · Team: Cremonese (R) · Pts: 34
```

CYCLE 2

THINK — Lecce is above. Now its coach.

ACT — ≈ Search by Meaning — `Lecce football club coach manager 2025–26`

OBSERVE — **D3** Lecce's Page

```
Ground:      Stadio Via del mare
Head coach:  Eusebio Di Francesco
```

ANSWER — ✓ **Eusebio Di Francesco** — **2** Tool Calls of 8 · Searched a Name It Just Read

THE STUDY

Three Research Questions

RQ1

Freshness

Does Retrieval Keep Answers Current When Facts Change, and at What Cost?

A Model Alone Stops at Its Training Text

1 Sep 2025 → 25 May 2026

S2 S3 S4 S5 vs S0

RQ2

Complexity

On Multi-Hop Questions, Does Searching Again Beat Searching Once?

A Pipeline Searches Once, an Agent Again

Cremonese → Lecce → Its Coach

S4 vs S2 S5 vs S3

RQ3

Architecture

Does an Agent Still Need a Vector Index, or Are Keywords Enough?

Claude Code Dropped Its Vector Index: “Agentic search outperformed everything.” (Boris Cherny)

`vector_search` or `keyword_search`

S4 vs S5

THE STUDY

The Judge: Claude Sonnet 5

QUESTION

As things stand, where are Hellas Verona in the Serie A standings?

ACCEPTED ANSWERS

`19th` `19` `nineteenth`

Written in Advance, for Each Question

WHY A JUDGE? **Exact Match Would Say No** `19th place (relegated)` ≠ `19th` Right, but Not Word for Word

S5'S ANSWER · 25 MAY 2026

`19th place (relegated)` →

Claude Sonnet 5
The Judge: Reads the Meaning

Sees the Question, Accepted Answers, Answer

"Is the answer to grade correct? The same person, team or number counts as correct even if written differently … Reply only yes or no."

✓ Yes

Same Place, Other Words

The Judge Is Not the Writer
Models Favour Their Own Answers

Writes the Answers (All but S1)
Claude Haiku 4.5

≠

Grades Them
Claude Sonnet 5

1,680 Answers Graded, Yes or No
7 Systems × 120 Questions × 2 Dates

THE STUDY

Can We Trust the Judge? A Second Opinion With No AI

THE SIMPLE RULE · NO AI · ON THE SAME ANSWER

Is an Accepted Answer Inside?

≠ Not Exact Match

S5	S5's Answer	`19th place (relegated)`	ACCEPTED ANSWERS: `19th` `19` `nineteenth`
1	Lowercase, No Accents	`19th place (relegated)`	Nothing Changes
2	Remove Punctuation	`19th place relegated`	`(` and `)` Disappear
3	Remove "a", "an", "the"	`19th place relegated`	None Here
4	Accepted Answer Inside?	`19th` `place relegated`	✓ Yes, "19th" — Extra Words Are Fine

THE JUDGE · AI

Claude Sonnet 5

✓ Yes — Reads the Meaning

= They Agree

SECOND OPINION · NO AI

The Simple Rule

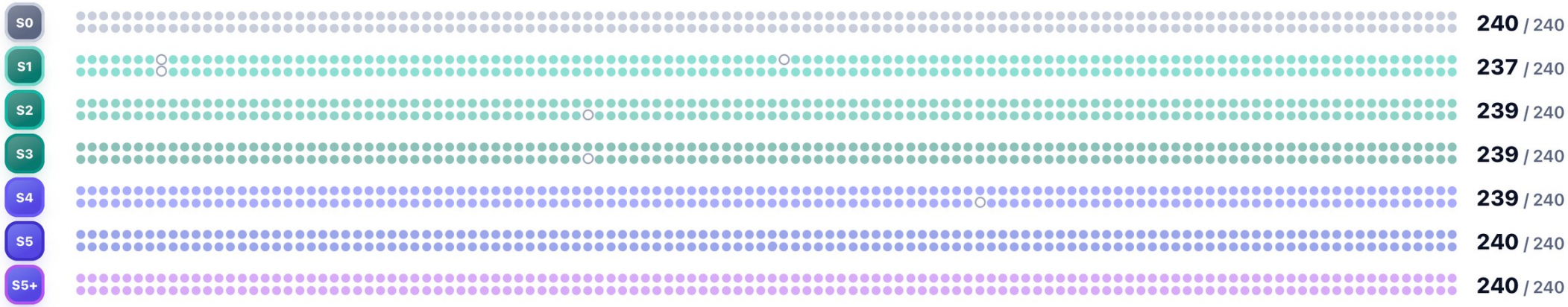
✓ Yes — "19th" Is Inside

THE STUDY

Can We Trust the Judge? A Second Opinion With No AI

ALL 1,680 ANSWERS · ONE DOT EACH 7 Systems × 2 Dates × 120 Questions

Agree Differ



RULE AND JUDGE AGREE

1,674

of 1,680 Answers

◉ 6 Differ 99.6%

THE 6 OTHERS

✕ ✕ **2 Errors of the Rule**

Sassuolo, “bottom place”: Right, but No “20th”

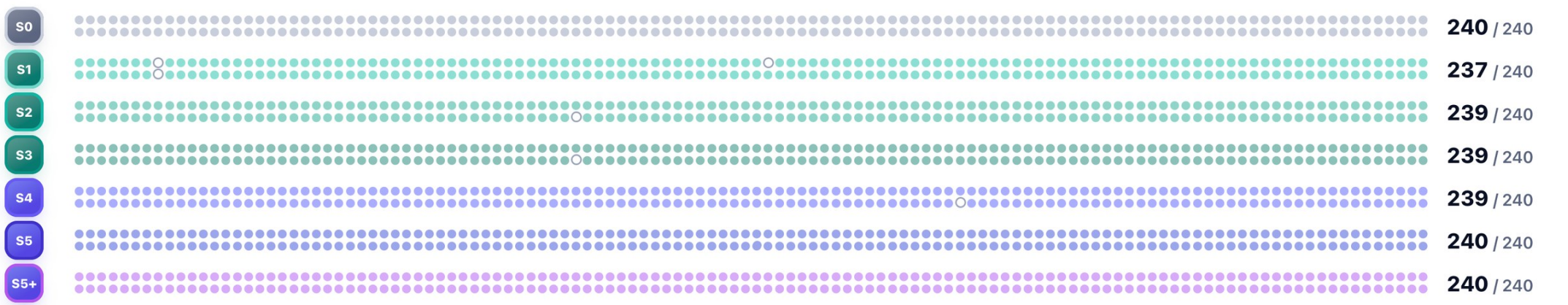
? ? ? ? **4 Debatable**

“Who is in charge at Juventus?” Coach or Executives?

THE STUDY

Can We Trust the Judge? A Second Opinion With No AI

ALL 1,680 ANSWERS · ONE DOT EACH 7 Systems × 2 Dates × 120 Questions



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Sassuolo, “bottom place”: Right, but No “20th”

? ? ? ? **4 Debatable**

“Who is in charge at Juventus?” Coach or Executives?

RE-GRADE EVERYTHING WITH THE RULE

The Same Conclusions

- ✓ No Accuracy Moves by More Than One Answer
- ✓ Every Comparison: Same Winner, Same Significance

PRACTICE

One Real Question, Seven Systems

Real Run · Question 82 · 25 May 2026

S0 S1 S2 S3 S4 S5 S5+

"Who is the manager of the club one place above Hellas Verona in Serie A at the moment?"

SERIE A TABLE · 25 MAY 2026

POS	CLUB	PTS
17	Lecce	38
18	Cremonese	34
19	Hellas Verona THE QUESTION'S CLUB	21

1 One Place Above
19th → 18th

2 Its Manager → Marco Giampaolo

PRACTICE

One Real Question, Seven Systems

Real Run · Question 82 · 25 May 2026

"Who is the manager of the club one place above Hellas Verona in Serie A at the moment?"

S0 × | S1 | S2 | S3 | S4 | S5 | S5+

S0 **S0** Closed Book

RIGHT ANSWER · MARCO GIAMPAOLO

1 Searches With

No Search
The Model Alone

2 The Library

993 Passages From 22 Pages — Hellas Verona's Page

NEEDED The Table: 18th, 19th

NEEDED Cremonese's Head Coach

Not Searched

3 Best 5 ×

NO PASSAGES

Haiku 4.5's Memory

Training Data Up to **July 2025**

4 Answer • LOGGED

Haiku 4.5
From Its Memory

I don't have access to current information about the ... Serie A season standings or current managers ... Answer: unknown

× **unknown** 1.2 s

× No Search: Its Memory Ends in July 2025, Before This Season Began.

PRACTICE

One Real Question, Seven Systems

Real Run · Question 82 · 25 May 2026

"Who is the manager of the club one place above Hellas Verona in Serie A at the moment?"

S0 × S1 × S2 S3 S4 S5 S5+

S1 S1 Classical NLP

RIGHT ANSWER · MARCO GIAMPAOLO

1 Searches With

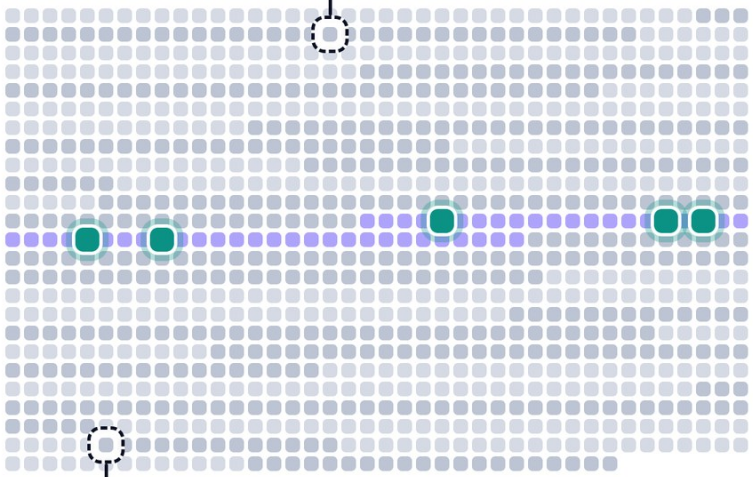
Aa **By Words**
BM25, on These Words

who · manager · club · one · place · above · hellas · verona · serie · moment

2 The Library

993 Passages From 22 Pages — Hellas Verona's Page

NEEDED The Table: 18th, 19th



NEEDED Cremonese's Head Coach

Scores **993** by Words · Keeps the **Best 5**

3 Best 5

RE-RUN

1. Hellas Verona FC — NOT HERE — Derby With Chievo Verona
2. Hellas Verona FC — NOT HERE — Rising Again and Serie A Comeback
3. Hellas Verona FC — NOT HERE — Supporters
4. Hellas Verona FC — NOT HERE — Rising Again and Serie A Comeback
5. Hellas Verona FC — NOT HERE — Origins and Early History

4 Answer

LOGGED

A Reader
Copies One Span

Its Best Span in Each of the 5
Weaker Than "Not Here"

Answer: unknown

× unknown 0.2 s

× None of Its 5 Holds the Answer, and a Span Is One Piece of One Passage: This Answer Needs Two.

PRACTICE

One Real Question, Seven Systems

Real Run · Question 82 · 25 May 2026

"Who is the manager of the club one place above Hellas Verona in Serie A at the moment?"

S0 × S1 × S2 × S3 S4 S5 S5+

S2 **S2** Naive RAG

RIGHT ANSWER · MARCO GIAMPAOLO

1 Searches With

≈ **By Meaning**
The Whole Question

THE QUESTION, AS WRITTEN

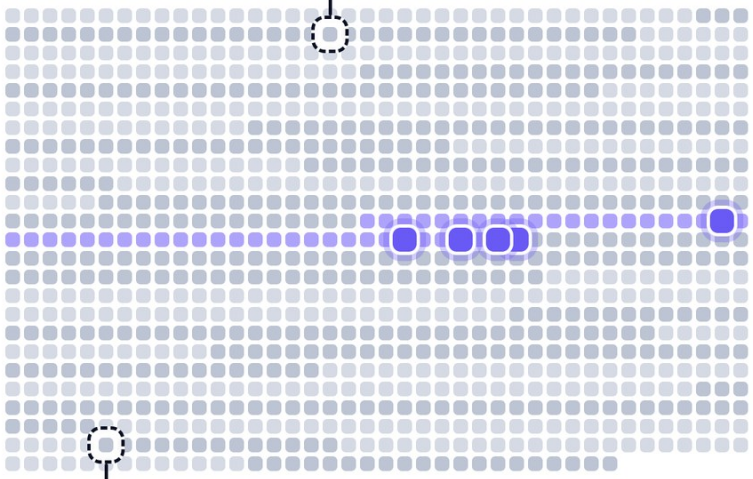
"Who is the manager of the club one place above Hellas Verona in Serie A at the moment?"

Cremonese Not in It

2 The Library

993 Passages From 22 Pages — Hellas Verona's Page

NEEDED The Table: 18th, 19th



NEEDED Cremonese's Head Coach

Scores **993** by Meaning · Keeps the **Best 5**

3 Best 5

RE-RUN

1. Hellas Verona FC — Managers
2. Hellas Verona FC — Managers
3. Hellas Verona FC — Rising Again and Serie A Comeback
4. Hellas Verona FC — Managers
5. Hellas Verona FC — Club Officials

4 Answer

LOGGED

Haiku 4.5
Writes From the 5

... the passages show that **Paolo Sammarco is the current manager of Hellas Verona** (as of 3 February 2026), they do not provide the league standings ...

× **unknown** 2.1 s

× The Question Names Verona, Never Cremonese: Five Verona Passages, No League Table.

PRACTICE

One Real Question, Seven Systems

Real Run · Question 82 · 25 May 2026

"Who is the manager of the club one place above Hellas Verona in Serie A at the moment?"

S0 × S1 × S2 × S3 × S4 S5 S5+

S3 **S3 Advanced RAG**

RIGHT ANSWER · MARCO GIAMPAOLO

1 Searches With

Aa ≈ **Words + Meaning**
The Whole Question

THE QUESTION, AS WRITTEN

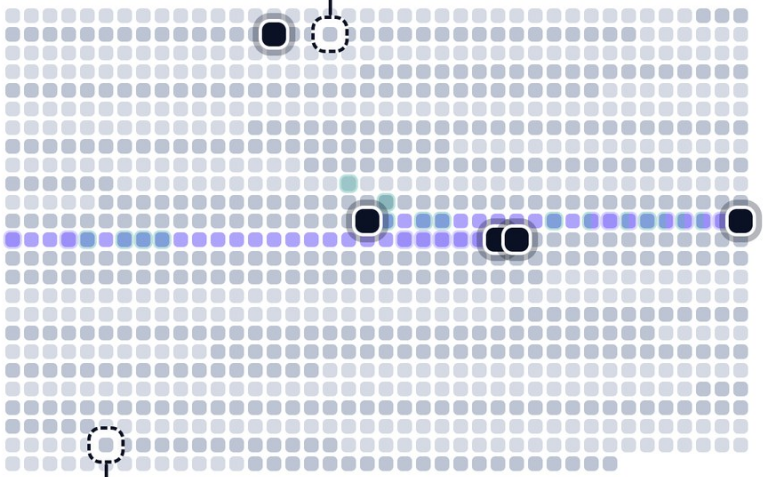
"Who is the manager of the club one place above Hellas Verona in Serie A at the moment?"

Cremonese Not in It

2 The Library

993 Passages From 22 Pages — Hellas Verona's Page

NEEDED The Table: 18th, 19th



NEEDED Cremonese's Head Coach

Best 20 by Words + Best 20 by Meaning

Fusion → 2025–26 Only → Re-ranker → **Best 5**

3 Best 5

RE-RUN

1. Hellas Verona FC — Managers
2. Hellas Verona FC — Managers
3. 2025–26 Serie A — Managerial Changes
4. Hellas Verona FC — Introduction
5. Hellas Verona FC — Rising Again and Serie A Comeback

4 Answer

LOGGED

Haiku 4.5
Writes From the 5

The passages only contain information about **Hellas Verona's managers** and some managerial changes in Serie A, but not the complete current standings ...

× **unknown** 6.1 s

× **Better Steps, the Same Question: Four of Five From Verona's Page, No League Table.**

PRACTICE

The Agents: Read a Name, Then Search It

Real Run · Question 82 · 25 May 2026

"Who is the manager of the club one place above Hellas Verona in Serie A at the moment?"

S0 × S1 × S2 × S3 × S4 ✓ S5 S5+

S4
Agentic RAG

ITS TOOLS
- ≈ Search by Meaning
- ▤ Read a Whole Page

TOOL CALLS · 8 AT MOST

≈ ≈

2 of 8 · Answered in 4.5 s

S5 · Agentic Search

S5+ · Agent + Tables

CALL 1

ACT ≈ Search by Meaning `Serie A standings 2025–26 current table` · LOGGED

OBSERVE — 1 of 5 Passages — 2025–26 Serie A League Table
`Pos: 18 · Team: Cremonese (R) · … · Pts: 34`
`Pos: 19 · Team: Hellas Verona (R) · … · Pts: 21`
↻ RE-RUN

LEARNS **Cremonese** Is 18th, Just Above Verona · ↑ INFERRED

CALL 2

ACT ≈ Search by Meaning `Cremonese manager 2025–26` · LOGGED

OBSERVE — 1 of 5 Passages — 2025–26 Serie A Managerial Changes
`Team: Cremonese · … · Incoming manager: Marco Giampaolo`
`Date of appointment: 18 March 2026`
↻ RE-RUN

ANSWER ✓ **Marco Giampaolo** "… Cremonese's manager is Marco Giampaolo, who was appointed on 18 March 2026 …" 2 Calls · 4.5 s · LOGGED

• LOGGED **Tool, Query, Answer, Seconds** — From the Run's Log

↻ RE-RUN **What the Tool Returned** — Not in the Log: Same Code, Query and Snapshot, Run Again

↑ INFERRED **What It Learns** — Read From Its Next Logged Step: a Query or the Answer

PRACTICE

The Agents: Read a Name, Then Search It

Real Run · Question 82 · 25 May 2026

"Who is the manager of the club one place above Hellas Verona in Serie A at the moment?"

S0 × S1 × S2 × S3 × S4 ✓ S5 ✓ S5+

S4 S4 · Agentic RAG 4.5 s
≈ ≈ Giampaolo

S5 S5
Agentic Search

ITS TOOLS
- Aa Search by Words
- Read a Whole Page

TOOL CALLS · 8 AT MOST
Aa Aa ▤ ▤ ▤ Aa

6 of 8 · Answered in 12.2 s

S5+ S5+ · Agent + Tables

CALLS 1–2

ACT Aa Search by Words `Serie A standings 2025–26 current` LOGGED

ACT Aa Search by Words `Serie A 2025–26 season table standings Hellas Verona position` LOGGED

LEARNS No League Table in Either Result ↑ INFERRED

CALLS 3–5

ACT ▤ Read a Whole Page `"2025–26 Serie A"` Part 1 Part 2 Part 3 LOGGED

LEARNS Cremonese Is 18th; Its Manager, Marco Giampaolo ↑ INFERRED

CALL 6

ACT Aa Search by Words `Cremonese manager 2025–26 Marco Giampaolo` LOGGED

OBSERVE 1 of 5 Passages
2025–26 Serie A
Personnel and Kits
`Team: Cremonese · … · Manager: Marco Giampaolo` ↻ RE-RUN

ANSWER ✓ Marco Giampaolo "… Cremonese is in 18th place … Marco Giampaolo is the manager of Cremonese …" 6 Calls · 12.2 s LOGGED

LOGGED **Tool, Query, Answer, Seconds**
From the Run's Log

↻ RE-RUN **What the Tool Returned**
Not in the Log: Same Code, Query and Snapshot, Run Again

↑ INFERRED **What It Learns**
Read From Its Next Logged Step: a Query or the Answer

PRACTICE

The Agents: Read a Name, Then Search It

Real Run · Question 82 · 25 May 2026

"Who is the manager of the club one place above Hellas Verona in Serie A at the moment?"

S0 ✗ S1 ✗ S2 ✗ S3 ✗ S4 ✓ S5 ✓ S5+

S4 · **S4 · Agentic RAG** 4.5 s
≈ ≈ Giampaolo

S5 · **S5 · Agentic Search** 12.2 s
Aa Aa ▤ ▤ ▤ Aa Giampaolo

S5+
S5+
Agent + Tables

ITS TOOLS
- Aa Search by Words
- ▤ Read a Whole Page
- ▦ Ask the Official Table

TOOL CALLS · 8 AT MOST
▦ | | | | | | | |

1 of 8

CALL 1

ACT ▦ **Ask the Official Table** `team "Hellas Verona" field "position" date "2026-05-25"` ● LOGGED

OBSERVE Written by the Engineer
Position, Points
From the Official Matches

↻ RE-RUN

`data/matches/matches.tsv`

matchday	date	home	away	home_goals	away_goals
1	2025-08-23	Genoa	Lecce	0	0
1	2025-08-23	Sassuolo	Napoli	0	2
⋮					
38	2026-05-24	Lecce	Genoa	1	0
38	2026-05-24	Torino	Juventus	2	2

… **380** Matches up to `2026-05-25`

TABLE ON 2026-05-25

Win 3
Draw 1
Loss 0
→
Points, Sorted

1–16	16 Clubs Above	
17	Lecce	38
18	Cremonese	34
19	**Verona**	**21**
20	AC Pisa	18

IT REPLIES

Verona is in position 19 on 2026-05-25.

LEARNS Verona Is 19th: the Club Above Is in Position 18 ↑ INFERRED

● LOGGED **Tool, Query, Answer, Seconds**
From the Run's Log

↻ RE-RUN **What the Tool Returned**
Not in the Log: Same Code, Query and Snapshot, Run Again

↑ INFERRED **What It Learns**
Read From Its Next Logged Step: a Query or the Answer

PRACTICE

The Agents: Read a Name, Then Search It

Real Run · Question 82 · 25 May 2026

"Who is the manager of the club one place above Hellas Verona in Serie A at the moment?"

S0	S1	S2	S3	S4	S5	S5+
×	×	×	×	✓	✓	✓

S4 · Agentic RAG 4.5 s
≈ ≈ ✓ Giampaolo

S5 · Agentic Search 12.2 s
Aa Aa ▤ ▤ ▤ Aa ✓ Giampaolo

S5+
Agent + Tables

ITS TOOLS
- Aa Search by Words
- ▤ Read a Whole Page
- ▦ Ask the Official Table

TOOL CALLS · 8 AT MOST
▦ Aa ▦

3 of 8 · Answered in 5.2 s

CALL 1
ACT ▦ Ask the Official Table `team "Hellas Verona" field "position" date "2026-05-25"` LOGGED
LEARNS Verona Is 19th: the Club Above Is in Position 18 ↑ INFERRED

CALL 2
ACT Aa Search by Words `Serie A standings 2025-26 season position 18` LOGGED
LEARNS Cremonese Is 18th ↑ INFERRED

CALL 3
ACT ▦ Ask the Official Table `team 'Cremonese' field "coach" date "2026-05-25"` LOGGED

OBSERVE
Written by the Engineer
Coach, Stadium
From the Club's Page
Its Infobox, 25 May 2026

US CREMONESE · INFOBOX

Ground:	Stadio Giovanni Zini
President:	Francesco Dini
Head coach:	Marco Giampaolo

→

IT REPLIES
Cremonese coach: Marco Giampaolo
↻ RE-RUN

ANSWER ✓ Marco Giampaolo "Answer: Marco Giampaolo" 3 Calls · 5.2 s LOGGED

• LOGGED **Tool, Query, Answer, Seconds** From the Run's Log

↻ RE-RUN **What the Tool Returned** Not in the Log: Same Code, Query and Snapshot, Run Again

↑ INFERRED **What It Learns** Read From Its Next Logged Step: a Query or the Answer

PRACTICE

Same Question, 1 September 2025

Real Run · Question 82 · 1 September 2025

"Who is the manager of the club one place above Hellas Verona in Serie A at the moment?"

S0	S1	S2	S3	S4	S5	S5+
×	×	×	×	×	×	×

S4 ON 1 SEPTEMBER 2025

Prompt: "Today is 2025-09-01. … check that your evidence is about the right season."

CALL 1

ACT ≈ Search by Meaning — Serie A standings 2024–25 current table — LOGGED

OBSERVE — 1 of 5 Passages — 2024–25 Serie A League Table
Pos: 13 · Team: Genoa · … · Pts: 43
Pos: 14 · Team: Hellas Verona · … · Pts: 37
RE-RUN

LEARNS — Genoa Is Just Above Verona: Last Season — INFERRED

CALL 2

ACT ≈ Search by Meaning — Genoa manager 2024–25 Serie A — LOGGED

OBSERVE — 1 of 5 Passages — 2024–25 Serie A Managerial Changes
Team: Genoa · … · Replaced by: Patrick Vieira
Date of appointment: 20 November 2024
RE-RUN

LEARNS — **Patrick Vieira**, Genoa's Manager in 2024–25 — INFERRED

ANSWER — × Patrick Vieira — "… for the 2024-25 Serie A season, Genoa's manager is Patrick Vieira …" — Right: Di Francesco — LOGGED

	25 MAY 2026	1 SEP 2025
S0	× unknown	× unknown
S1	× unknown	× unknown
S2	× unknown	× unknown
S3	× unknown	× unknown
S4	✓ Marco Giampaolo	× Patrick Vieira
S5	✓ Marco Giampaolo	× Patrick Vieira
S5+	✓ Marco Giampaolo	× Patrick Vieira
	Right: Giampaolo Cremonese, 18th	Right: Di Francesco Lecce, 17th

Genoa's Coach: in Last Season's Table, Genoa Was One Place Above Verona.

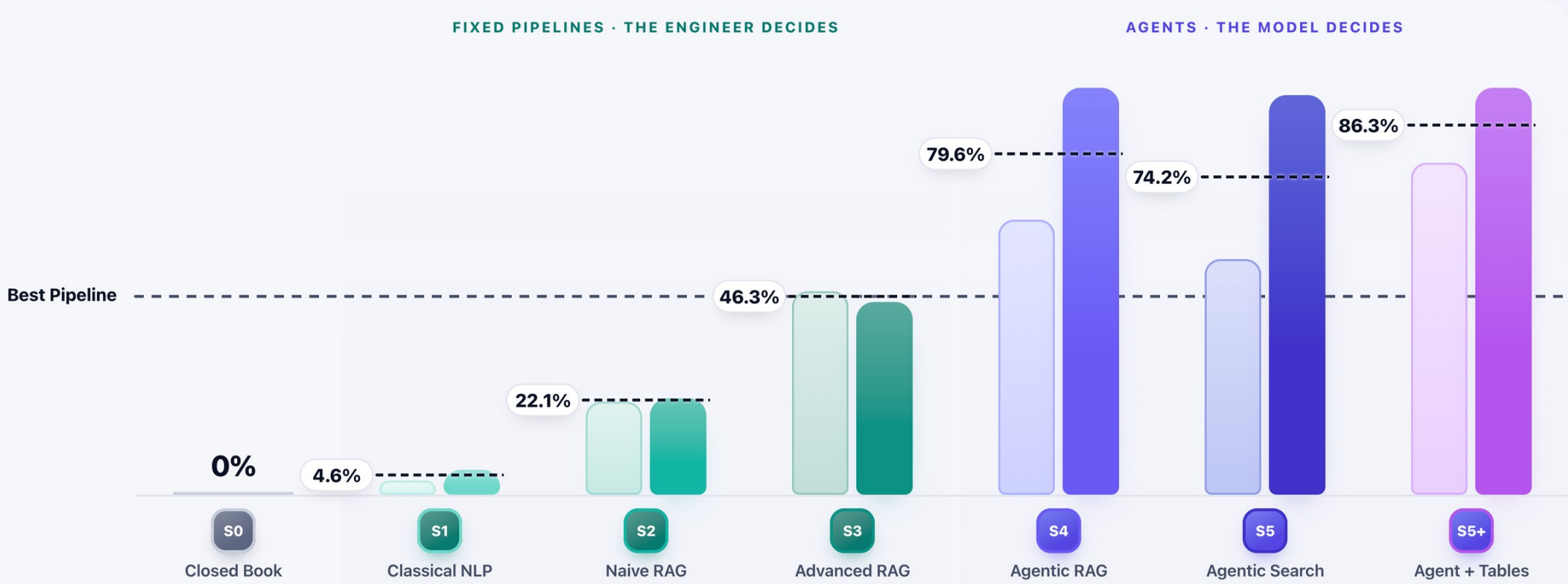
The Same Template for Five Clubs · 25 May: S0–S3 **0 of 5**, Every Agent **5 of 5** · 1 September: S4 **2**, S5 **1**, S5+ **1**. Right Moves, **Wrong Season**.

RESULTS

Handing the Search to the Model Pays Off

Static 40 · Dynamic 40 · Multi-Hop 40 — All **120 Questions × 2 Dates**: 240 Answers per System

1 Sep 2025 · 25 May 2026 · Dashed: Their Mean



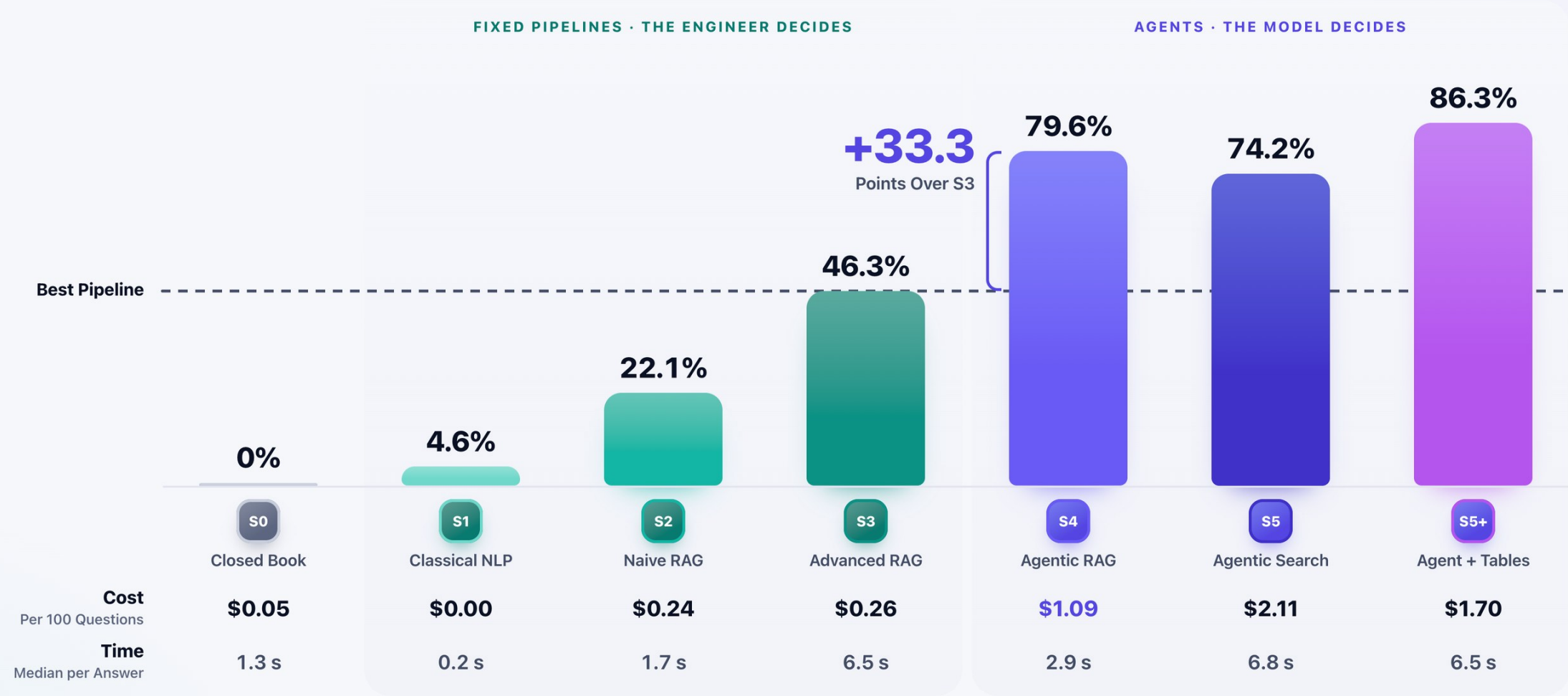
	S0 Closed Book	S1 Classical NLP	S2 Naive RAG	S3 Advanced RAG	S4 Agentic RAG	S5 Agentic Search	S5+ Agent + Tables
1 Sep 2025	0%	3.3%	21.7%	47.5%	64.2%	55.0%	77.5%
25 May 2026	0%	5.8%	22.5%	45.0%	95.0%	93.3%	95.0%

RESULTS

Handing the Search to the Model Pays Off

Static 40 · Dynamic 40 · Multi-Hop 40 — All **120 Questions × 2 Dates**: 240 Answers per System

Each Bar: The Mean of **1 Sep 2025** and **25 May 2026**



RESULTS

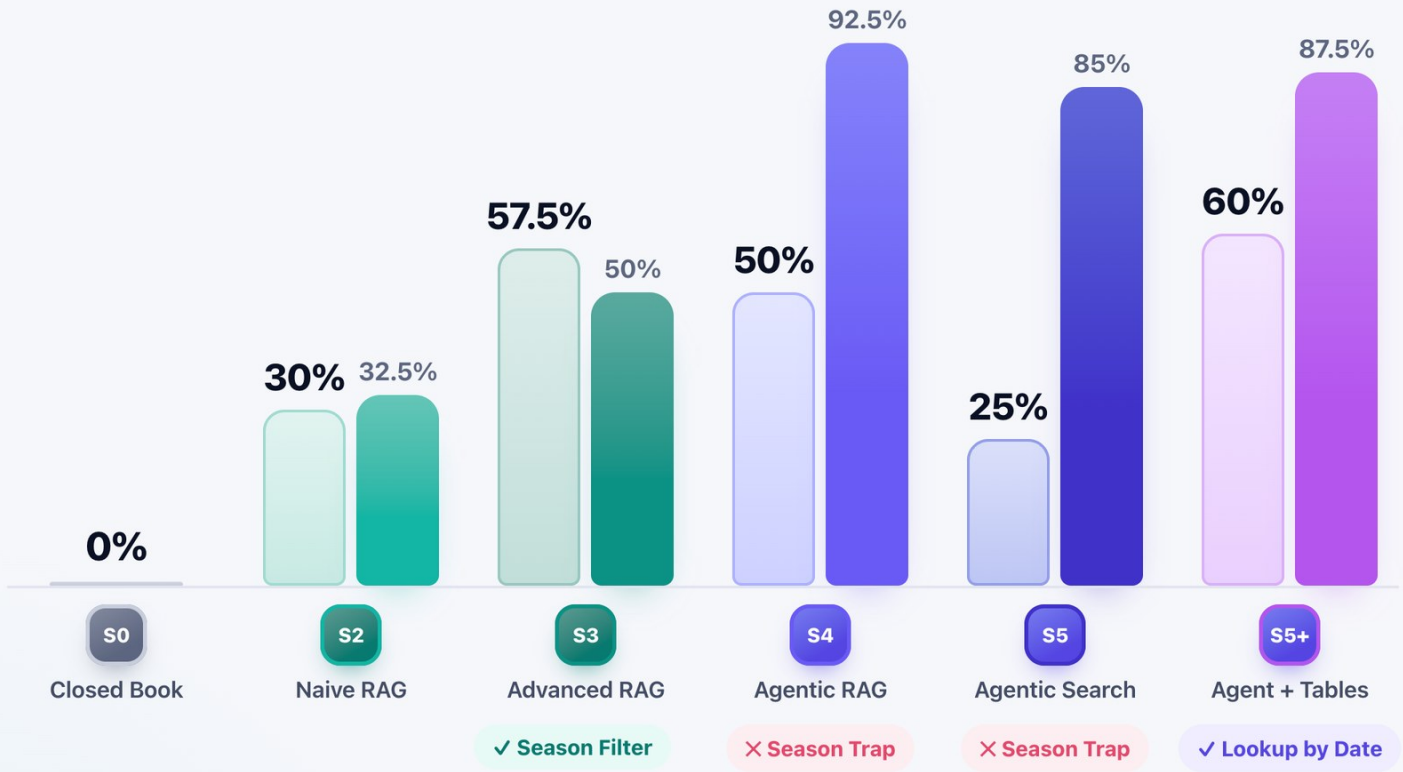
RQ1 · Freshness, and the Season Trap

Static 40 | Dynamic 40 | Multi-Hop 40

Only the **40 Dynamic Questions**, About Now, on Each Date

1 Sep 2025 The Start of the Season

25 May 2026



KEEPING THE INDEX CURRENT · 25 MAY

Time to Rebuild 993 Passages

Keyword Index	0.04 s
Vector Index	21.48 s

THE SEASON TRAP · 1 SEP 2025

"...how many league points Napoli have picked up this season?"

This Season 2025–26	6 pts	S5+ ✓
Last Season 2024–25	82 pts	S2 S4 S5 ✕

S4 S5 "Napoli Serie A 2024–25 season points"

First Search Named "2024" · This Season's 67 Questions

S4 59 / 67 S5 60 / 67

RESULTS

RQ2 · Complexity: Read the Middle Club, Then Search It

Static 40 | Dynamic 40 | Multi-Hop 40

Only the **40 Multi-Hop Questions × 2 Dates**: 80 Answers per System

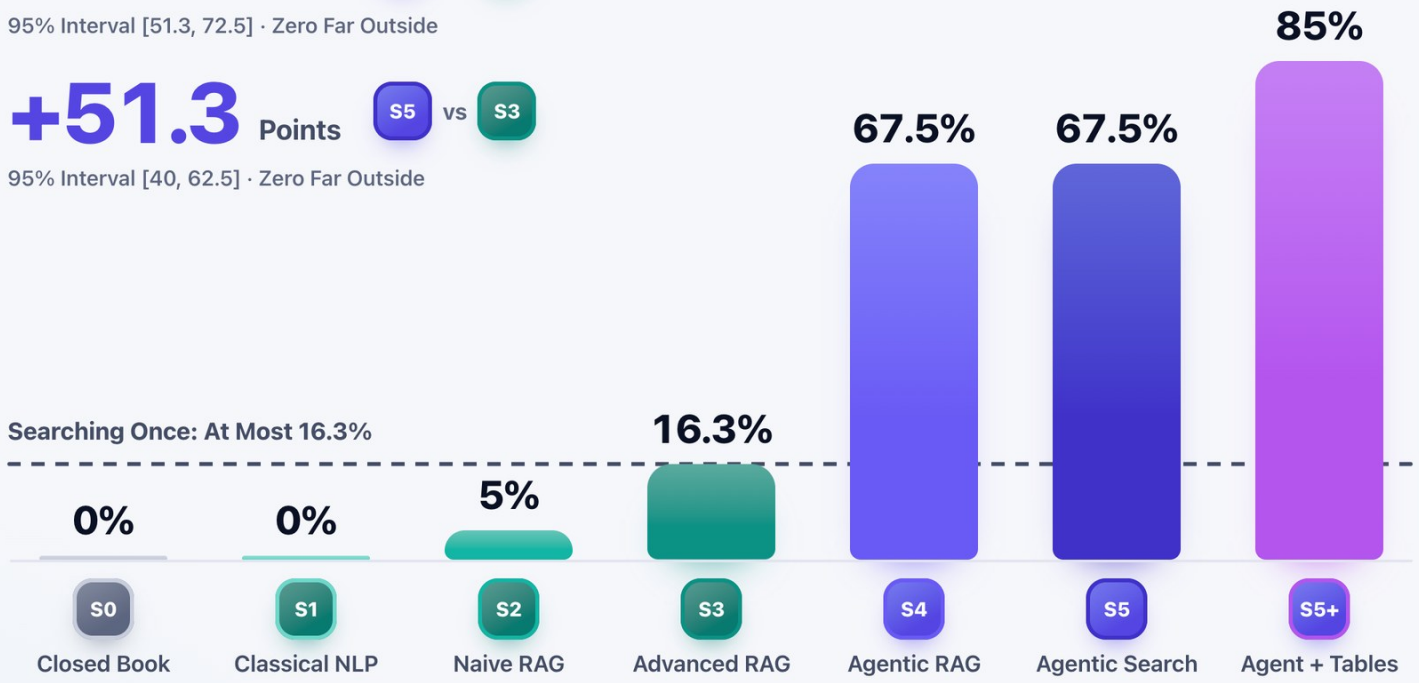
SEARCHING AGAIN VS SEARCHING ONCE

+62.5 Points S4 vs S2

95% Interval [51.3, 72.5] · Zero Far Outside

+51.3 Points S5 vs S3

95% Interval [40, 62.5] · Zero Far Outside



RESULTS

RQ2 · Complexity: Read the Middle Club, Then Search It

Static 40 | Dynamic 40 | Multi-Hop 40 — Only the **40 Multi-Hop Questions × 2 Dates**: 80 Answers per System

WHY SEARCHING ONCE FAILS

One Search Can Find the Middle Club, Not the Answer

A Bridge Question
30 of the 40

The Club Named → Search 1 → The Middle Club → Search 2 → The Answer

Stopped at the Middle Club
Wrong, but Naming It

S2 18 | S3 17 | S4 4 | S5 5
of 60 Bridge Answers Each

S4's Real Run
Question 82 · 25 May

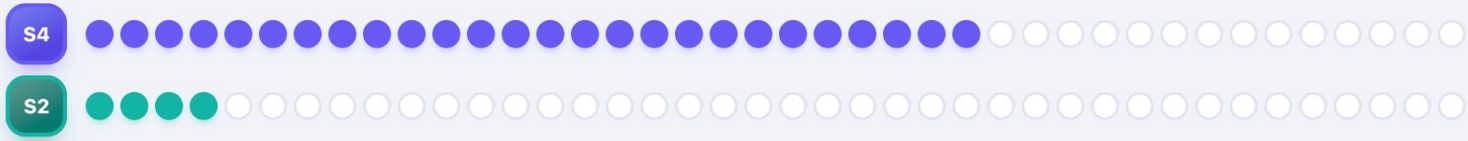
Hellas Verona — Cremonese — Marco Giampaolo ✓

RIGHT ANSWERS	S2	S3
Every Club Named 10 Questions × 2 Dates	15% 3 of 20	30% 6 of 20
A Middle Club First 30 Questions × 2 Dates	2% 1 of 60	12% 7 of 60

WHAT HELPS THE AGENTS · S4'S 80 MULTI-HOP ANSWERS, S2 ON THE SAME QUESTIONS

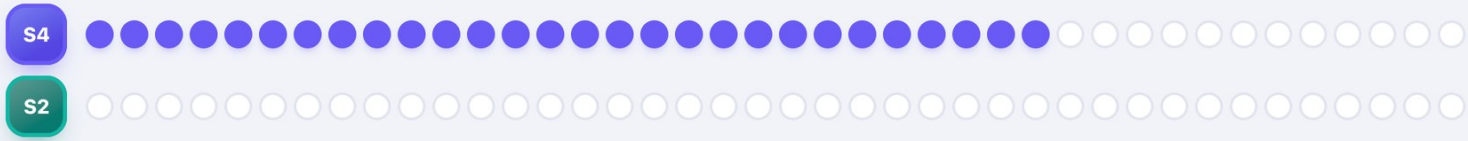
Two Things Help: Its Own Query, Then Searching Again

1 · One Search
Its Own Query
S2 Uses the Question as Written



26 / 40
4 / 40

2 · Two or More Searches
Searching Again
S2 Never Searches Twice



28 / 40
0 / 40

RESULTS

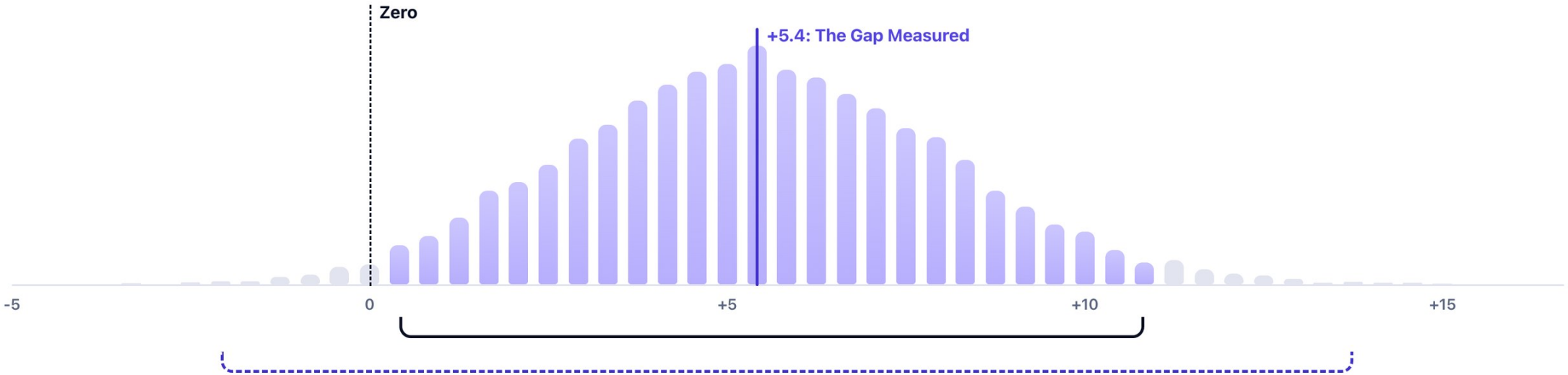
RQ3 · The Index Buys Lower Cost and Speed, Not Accuracy

The Same Agent, With a Vector Index (S4) or Keyword Search Only (S5)

RQ3 · ACCURACY · A PAIRED BOOTSTRAP

Is S4's 5.4-Point Lead Real?

1 Draw 120 Questions at Random → 2 Score S4 and S5 on the Same Draw → 3 Repeat 10,000 Times



95% Interval [+0.4, +10.8] Zero Outside → Significant

Bonferroni, 15 Comparisons [−2.1, +13.8] Zero Inside → Not Significant → S4 ≈ S5 No Proven Accuracy Gap

RESULTS

RQ3 · The Index Buys Lower Cost and Speed, Not Accuracy

The Same Agent, With a Vector Index (S4) or Keyword Search Only (S5)

S4 ≈ S5 **No Proven Accuracy Gap**

Why No Gap? Agents Write the Pages' Words

Meaning Search Helps Mostly With Synonyms
"In Charge" or "Manager": Same Meaning

→

But Agents Write Their Own Query
In the Words the Pages Use

→

So Keywords Find the Page Too
S5 Keeps Up Without a Vector Index

QUESTION 44 · 25 MAY 2026

~~Who's~~ ~~in charge~~ at Pisa ~~at the moment~~?

THE PAGES SAY "MANAGER"

→

S4 AND S5 BOTH SEARCHED

"Pisa football club **manager** 2026"

✓ Oscar Hiljemark

0 of 678
Agent First Queries Copied the Question

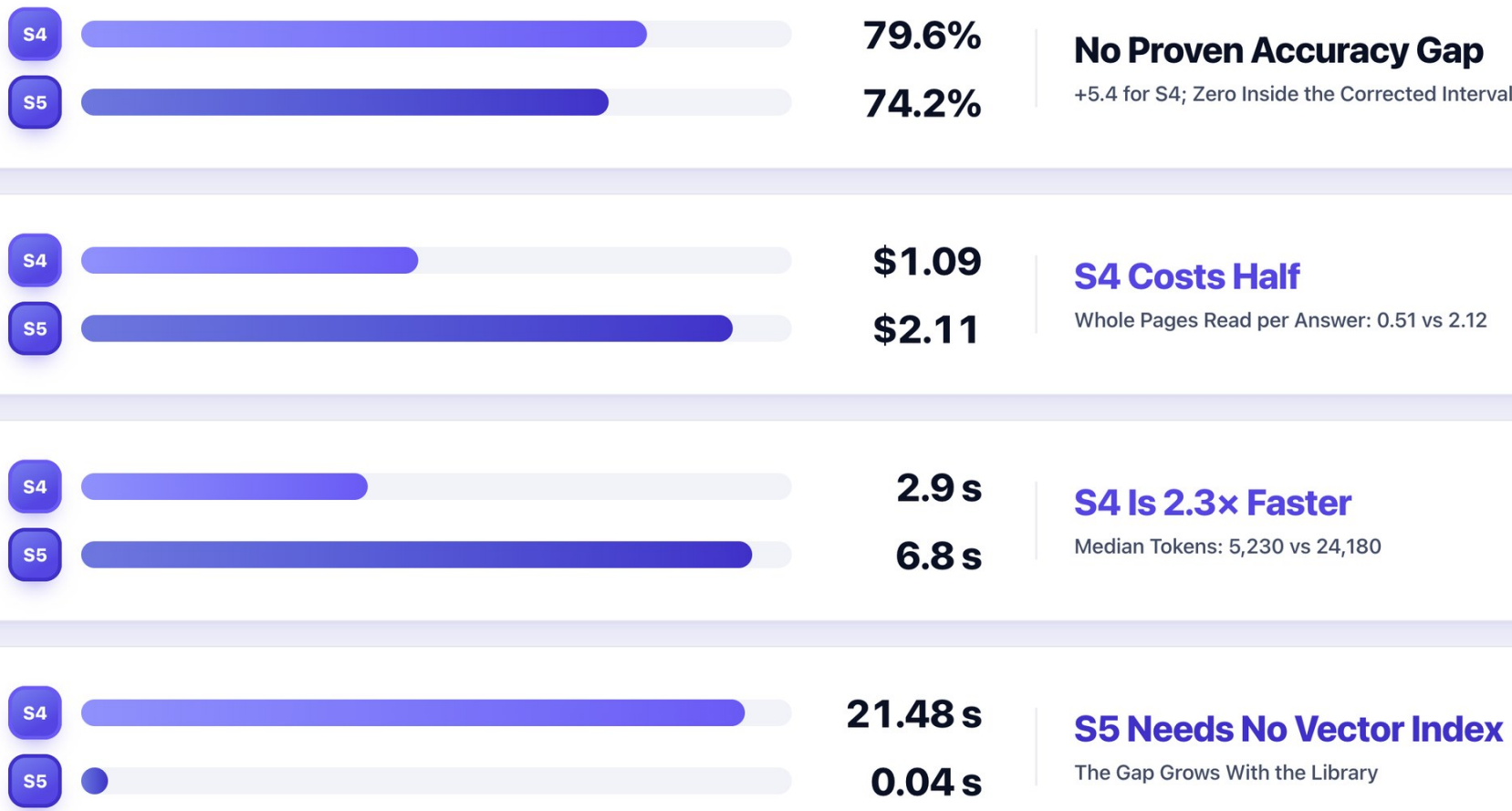
6 vs 13
Median Words: First Query vs Question

90% Right
Coach Questions With Synonyms, S4 and S5

RESULTS

RQ3 · The Index Buys Lower Cost and Speed, Not Accuracy

The Same Agent, With a Vector Index (S4) or Keyword Search Only (S5)



CLOSE

No Single Best System: One for Each Need

IF YOU NEED	PICK	BECAUSE
Facts That Changed Since Training	S2 S3 S4 S5 S5+ **Retrieval + an LLM** — Never the Model Alone	S0 **0%** The Model Alone, on Both Dates
Single Facts on a Tight Budget	S3 **S3 · Advanced RAG** — One Search, the Cheapest LLM System per Right Answer	**61%** of Single-Fact Questions, Both Dates
Questions That Chain Two Facts	S4 **An Agent · S4, the Cheapest One** — It Reads the Middle Club, Then Searches It	**68%** on Multi-Hop · Pipelines at Most 16%
A Huge or Often Rebuilt Library	S5 **S5 · Keywords Only** — No Vector Index to Rebuild · Twice S4's Cost	**0.04 s** to Rebuild, Against 21.48 s
An Official Table Exists	S5+ **Give It to the Agent · S5+** — Its Lookup by Date Finds the Right Season	**86%** Overall · +12.1 Points Over S5
Old and New Facts Side by Side	S3 or S5+ **Season Filter or Date Lookup** — The Date in the Prompt Was Not Enough	1 Sep 2025 · Questions About Now — S3 **57.5%** S5+ **60%** S5 **25%**

CLOSE

The Limits: What These Results Cannot Tell

Template Questions

1

120 Short, Factual Questions
From 17 Templates, in English

SO
No Vague Questions, Where
Meaning Search Should Help Most

One League, One Model

2

Serie A on English Wikipedia;
Every Answer by Claude Haiku 4.5

SO
Another Model May Fall Into
the Season Trap Less Often

A Small Library

3

Under 1,000 Passages per Date

SO
With Millions, a Vector Index
Takes Far Longer to Rebuild

A Single Run

4

Each System Answered Each Question Once

SO
Asked Again, S5 Moved 5
Points: About the S4–S5 Gap

S3 Tuned on the Test

5

Re-Ranker Chosen After 10 Test
Questions; Season Filter Written for Them

SO
On New Questions, S3 May Do Worse

Three Changes at Once

6

An Agent Writes Its Query, Searches
Again and Reads Whole Pages

SO
Which of the Three Helps Most Is Not Tested

CLOSE

Back to the Opening Question

"Who coaches the team directly above Cremonese in the Serie A table right now?"

Cremonese → Lecce → Eusebio Di Francesco

25 May 2026 · On 1 Sep: Napoli → Antonio Conte

1

The Latest Documents

+25 to **+93**

RQ1 · Points Over the Model Alone

2

Two Searches

+51 to **+63**

RQ2 · Points Over Searching Once

3

Knowing Which Season It Is

S4 **37** of 42 S5 **46** of 50

1 Sep · Wrong Answers From Last Season

Thank You.

Interactive Page · alexcambier.github.io/seriea-agentic-search

Code and Data · github.com/AlexCambier/seriea-agentic-search